Management of Change: BASF process column K 2750 (DEXPI C02, unseen)

DRAFT - NOT APPROVED. Prepared 2026-11-02 14:46 +03 by PlantAgent from the working model. Nothing in this file is an approval.

1. Summary

Decision requested	**Approve for detailed engineering: 1 change to reach 120 % of today's throughput - (1) Replace column with 600 mm ID rated shell (vendor to confirm).**
Target	120 % of today's throughput
Result (DWSIM, re-solved for this file)	At 120 %, existing plant: 1 limit over 100 % (highest K 2750 distillation Fair flood fraction vs limit 110 %); with the changes: all modelled process limits within 100 % (highest K 2750 distillation Fair flood fraction vs limit 76 %). Simulated maximum throughput: 113.1 % → 145.1 % of today.
Cost class (agent estimate, unpriced)	(1) not stated
Needs approval	(1) Replace column with 600 mm ID rated shell (vendor to confirm): cost class not stated (agent estimate), replacement rating (vendor confirmation open); Pre-startup safety review: 6 items (section 6.2).
Status	DRAFT for engineering review; to be checked by a licensed engineer. Sign-off in section 7.

Agent's explanation on record (model text, quoted; its figures are not re-checked here: sections 3 and 4 are the calculation record): Flood limits first - 600 mm shell needed for envelope. K 2750 (ProcessColumn-1: 500 mm ID, 14 000 mm, 6 bar / 150 °C) is tight at +20 % feed and fails the +10 % operating envelope. With bottoms held at 3.0 mol/s flood is 0.79 vs 0.80 = 98.8 % at 1.2× and 1.008 vs 0.80 = 126 % at 1.32× (K2750_1p2_scaled_500, K2750_1p32_fixed_500) and base split is lost. Ratio-controlled bottoms 3.0→3.6 mol/s at 1.2× (3.96 at 1.32×) holds purity and 600 mm ID gives 0.604 vs 0.80 = 75.5 % at 1.32× (K2750_1p32_600_verify) with +20 % duty (+32 % at envelope) and all documented P/T checks < 61 %.

2. Basis

Question. +20 % feed: what limits it first and what does it cost?

Plant and data sources. Uploaded by the engineer.

Data status. The drawing (official DEXPI example C02, unmodified) shows the column but not its service: Muse assumed the duty (water/ethanol, 1,000 kg/h, 85 °C, 3 bar). Results hold for that assumed duty, not BASF's real one. The purity spec is a draft (today's purity within 0.5 points) until an engineer sets one.

Model. K2750_submitted built by baseten:meta-models/muse-spark-open-1026 on 2026-11-01 20:35; simulator DWSIM (headless). Review status: **ok** (0 findings, 0 blocking; details in appendix C).

3. Existing vs proposed at 120 % (DWSIM, re-solved for this file)

Item	Check	Limit	Existing	Proposed	Proposed value
K 2750 distillation	Fair flood fraction vs limit	0.8 fraction of flood	110 %	**76 %**	0.61 fraction of flood

Existing and Proposed are utilisation (value / limit) of each limiting item at the target; over 100 % fails. For an energy-closure check the limit is the heat available in that solve.

4. Changes

4.1 Replace column with 600 mm ID rated shell (vendor to confirm)

Cloud 1 on the Rev B drawing. Cost class (agent estimate, unpriced): not stated.

What physically changes.

- Replaces K 2750 distillation: diameter m 0.6 (DWSIM Fair flood requires 0.638 m at 1.32× scaled 3.96 mol/s; 600 mm uniform (next standard matching KX 2750 600 mm section) gives 0.604 vs 0.80 =75.5 % at envelope (K2750_1p32_600_verify) and 0.549 =68.6 % at 1.2× (K2750_1p2_600 family); 525 mm gives 0.789 =98.6 % no margin (K2750_1p32_525). Height 14000 mm, 6 bar/150 °C, 1.4571, 15 stages same.; vendor to confirm)
- On the drawing: K 2750 REPLACED: diameter m 0.6 (vendor to confirm).

Why. Existing 500 mm ID cannot pass the +10 % operating envelope even with correct split: 0.869 vs 0.80 =108.6 % at 1.32× scaled (K2750_1p32_scaled_500) and 1.008 =126 % fixed (K2750_1p32_fixed_500). All documented P/T checks pass (<61 % at 1.32×) so flooding and purity are the only documented limits. New 600 mm ID provides required capacity with margin; vendor to confirm tray/valve design, downcomers, nozzles for line 2750.001.6 and *at N3*/*C2* valve.

Rating and rating source.

- K 2750: agent-proposed sizing basis (not independently validated): DWSIM Fair flood requires 0.638 m at 1.32× scaled 3.96 mol/s; 600 mm uniform (next standard matching KX 2750 600 mm section) gives 0.604 vs 0.80 =75.5 % at envelope (K2750_1p32_600_verify) and 0.549 =68.6 % at 1.2× (K2750_1p2_600 family); 525 mm gives 0.789 =98.6 % no margin (K2750_1p32_525). Height 14000 mm, 6 bar/150 °C, 1.4571, 15 stages same.; vendor confirmation open

Before / **Proposed calculation** (this change alone, DWSIM at the target).

- Simulated maximum: 113.1 % → 145.1 % of today; first limit K 2750 → K 2750.

Item	Check	Limit	Before	After
K 2750	Fair flood fraction vs limit	0.8 fraction of flood	0.88 fraction of flood (110 %)	0.61 fraction of flood (76 %)

Field checks that could change this.

- Actual product purity spec for 2750.001.6 and *at N3* vent/distillate. Where: BASF process datasheet/DCS lab specs for K 2750 - confirm bottoms and distillate targets that define if fixed 3.0 mol/s is off-spec. If it differs: If spec is tighter than base split, required reflux or stages may increase and flood/duty numbers change; if looser, fixed bottoms could be tolerated at 1.2× but still fails flood at 1.32×

4.2 Other field checks

- Feed rate/composition envelope and actual reboiler/condenser capacity. Where: DCS historian for FEED flow, temperature and lab composition; nameplates for reboiler steam valve and condenser - measure max duties available. If it differs: If available duty <563 kW the 1.32× envelope cannot be met even with new shell - would need utility upgrade
- Tray type, spacing and fouling/pressure-drop. Where: Column internals drawing and manway inspection - confirm sieve/valve trays and actual 0.5 m spacing assumed. If it differs: Different spacing or high-pressure-drop internals shifts flood correlation and the 0.638 m required diameter

4.3 Economics and uncertainty of the design

Economics (ILLUSTRATIVE prices; order of magnitude)

- draft (assumed values): Capex ≈ $0k (class 5: $0k–$0k); product price not given: margin, payback and NPV open (energy and utilities $0k/yr). All prices ILLUSTRATIVE.

Proposal	Stated class	Factored capex	Check
Replace column with 600 mm ID rated shell (vendor to confirm)		$0	

- Added Process feed: 5 t/d × 330 d; product margin $0/yr (solved 1.451 × base, capped at the 120% target; energy from DWSIM solves (design at 120% vs today at 100 %)).
- Outage for tie-ins/cleaning: 0 d → $0 (muse_default shutdown for a tie-in or exchanger clean).
- Payback None years; NPV $0 (capex band: $0 to $0).

Uncertainty of the design (Monte Carlo over assumed and illustrative inputs)

- Target 120% met in 100% of 48 samples; max throughput P10/P50/P90 144% / 145% / 145%; most often limited by K 2750: Fair flood fraction vs limit (100%); biggest driver feeds.FEED.T_C (assumed).
- Method: Latin hypercube, 48 samples (seed 20261101), uniform ± band per input kind; each sample re-solved (DWSIM): capacity bisection plus the solve at the target; 48 of 48 samples solved.

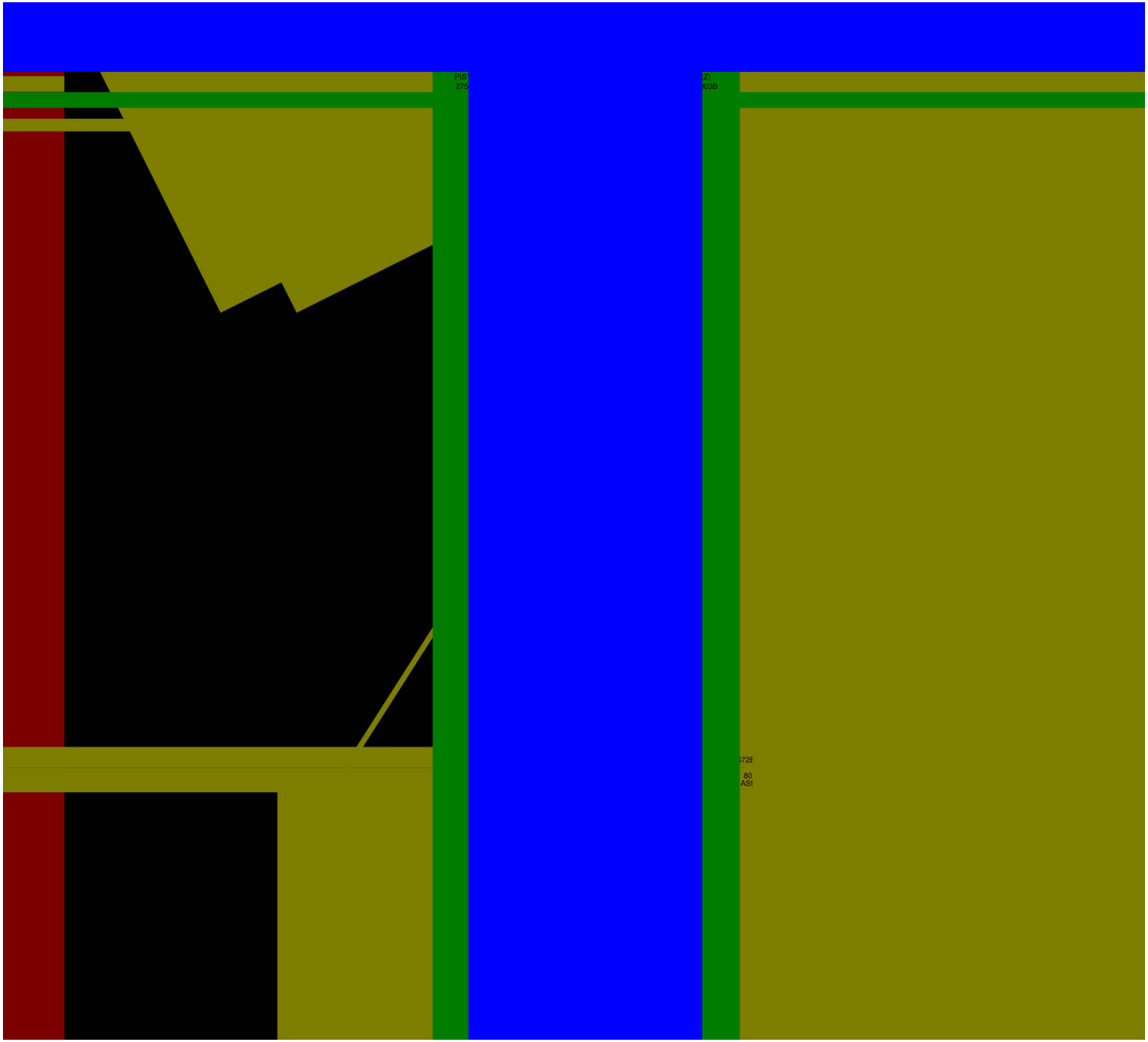
Input	Kind	Base	± band
units.*at N3*/*C2*.dP_bar	assumed	0.487	10 %
units.K 2750.dP_bar	assumed	0.03	10 %
feeds.FEED.P_bar	assumed	3.0	10 %
feeds.FEED.T_C	assumed	85	10 %
limits.*at N3*.max_T_C	assumed	150	10 %
limits.FEED.max_T_C	assumed	150	10 %
limits.VENT_INT.max_T_C	assumed	150	10 %

- K 2750: Fair flood fraction vs limit at the target: load P10/P50/P90 76% / 76% / 77%.

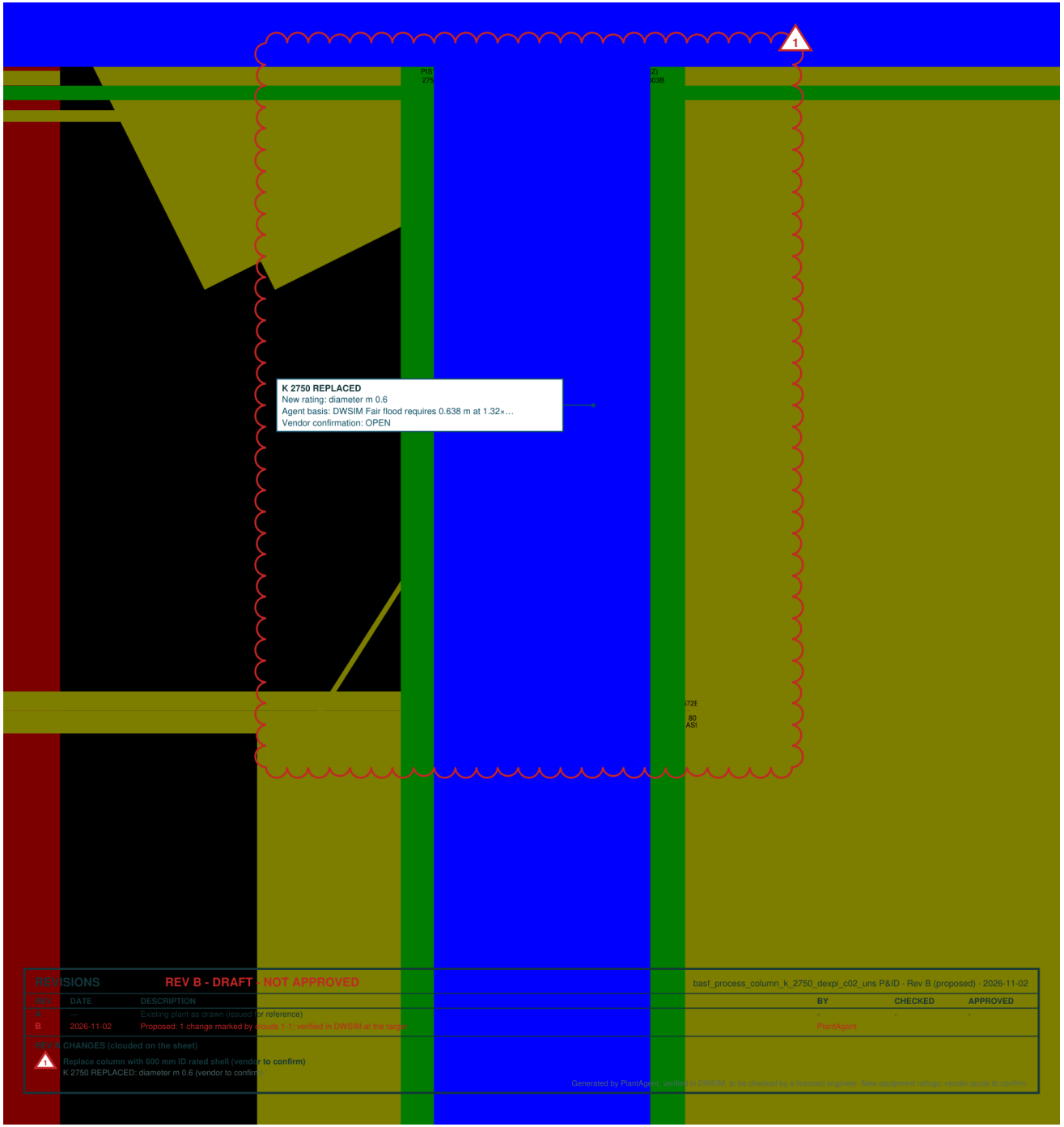
5. Drawings

The next pages show the P&ID **Rev A** (existing, as received) and **Rev B** (proposed): each change is drawn in, clouded and numbered as in section 4, with the new design values written next to the equipment or line and a revision block (DRAFT - NOT APPROVED). Files in the pack: `rev_a.svg`, `rev_b.svg`, `rev_b.png`.

5.1 P&ID Rev A - existing (as received)



5.2 P&ID Rev B - proposed (DRAFT - NOT APPROVED)



6. Risks and safety

6.1 Relief protection (P&ID relief scan)

- **Finding:** K 2750 all side: no relief on this P&ID. no relief device reachable (lines leave the drawing: check the connecting P&ID).

6.1c Discipline gates (outside the process model)

Pass/fail only from a sourced check; a check over an ILLUSTRATIVE rating leaves its gate open until the rating is confirmed. Electrical is in section 6.1b.

Gate	Status	Summary
robustness	**pass**	draft (assumed values): operating envelope 2/2 scenarios pass
piping utilities	**not affected**	no physical or routing change
controls	**not affected**	no physical or routing change
process safety	**not affected**	no physical or routing change
civil structural	**not affected**	no physical or routing change

6.2 Pre-startup safety review (PSSR), generated from the changes

- [ ] Replacement of K 2750 distillation: confirm the installed rating (diameter_m = 0.6) against the vendor data sheet before start-up.
- [ ] Mechanical and process safety review for replacement K 2750: confirm materials, nozzles, supports, alarms, trips and relief basis are suitable for the new rating.
- [ ] Commissioning of replacement K 2750: update the equipment data sheet and procedures, then record the first operating point against the model.
- [ ] Update the P&ID, equipment datasheets and the operating procedure for every change above.
- [ ] Brief the operators on the new setpoints before start-up.
- [ ] Record the first 24 h of operation at the new throughput and compare with the model's predicted values in this file.

6.3 Second opinion

No second opinion was requested for this answer.

6.4 Open risks

- 1 relief finding on changed equipment (section 6.1).

7. Approvals (DRAFT: not approved)

Role	Name	Signature	Date
Process engineer			
Operations			
Process safety (MOC)			
Manager (approver)			

Appendix A. Reproducibility

- Code version (git): c2b7100; simulator: DWSIM (headless) via plantagent.process.
- Audit-chain head hash: `c5ab20c8b4bd3d70718d6d305703d5a4a18b8f9ef713c3d02877f0ee5987716c` (chain intact: True).
- Model: K2750_submitted by baseten:meta-models/muse-spark-open-1026 on 2026-11-01 20:35; answer by baseten:meta-models/muse-spark-open-1026.
- Run labels and calculation records: K2750_submitted.
- Sections 1, 3 and 6.2 were recomputed when this file was made; re-run `python -m plantagent.decision basf_process_column_k_2750_dexpi_c02_uns` on the same revisions to reproduce them.

Appendix B. Capability gaps

- None recorded.

Appendix C. Model review

Status: **ok**.

- No findings.

An overall energy balance is not closed: the simulator's streams do not report enthalpy. Unit duties are checked one by one instead.

Appendix D. Model assumptions

Recorded by the model builder when the model was built (model text). Later model corrections (section 2) take precedence; items the current model no longer matches are marked **superseded**.

- Fluid composition assumption: Line 2750.001.6 fluid code 8503 and piping class 10HB25T10B1 not defined in project documents (search_documents for K2750/C02/8503/10HB25T10B1/BASF/DEXPI returned no pages). Data sheet states no process conditions on drawing. Model uses Water+Ethanol with NRTL as generic organic-water proxy to demonstrate column hydraulics; all temperatures/duties/compositions illustrative and must be replaced with actual fluid 8503 assay.
- Operating condition assumption: No feed temperature, pressure, flow or product specs on P&ID or data sheet. Assumed feed 1000 kg/h, 85 C, 3 barg, x Water 0.3 / Ethanol 0.7; reflux ratio 1.5, reboiler product 3.0 mol/s; condenser total. No measured values to calibrate against - calibration section reports design-limit utilisation only.
- Column staging assumption: Height 14000 mm with tray spacing 0.5 m allows ~28 theoretical trays max. Model uses 15 stages feed stage 8 efficiency 0.7 for numerical stability and to fit within shell; actual internals (trays vs packing, stage count) not on DEXPI export.
- Diameter conflict/assumption: K 2750 Chamber 1 InsideDiameter 500 mm vs KX 2750 ColumnSection InsideDiameter 600 mm - both Material 1.4571. Interpreted as single column with two diameters (enlarged section). Model uses conservative 0.5 m for flood screening; DWSIM estimates 0.555 m required at assumed rate, indicating tight fit - to be checked once actual vapour loads known. Listed as assumption per instruction to note site/P&ID conflicts.
- Exclusions justified: Valves *at P3*/OperatedValve and *at P4*/OperatedValve and lines *at P3*/*at P4* are instrument root valves to BlindFlange -> PI(Z) 275003A/B -> PIS+Z+A 275003 pressure measurement loops, not process flow lines; no connection to off-page flow. Closed per check_coverage guidance and not modelled as process valves. No data_quality warnings present on latest P&ID export.
- Topology assumption: Recirculation/utility loops: none on this DEXPI C02 extract. Both documented connections carry normal flow and are modelled: model stream VENT_INT via *at N3*/*C2* to open-out (*at N3*) and bottoms to 2750.001.6 off-page connector via PipeReducer. No recycle unit needed (no temperature-by-mixing loop; no closed coolant loop at known supply temperature).
- Limits assumption: All limits taken from P&ID design_limits (max_P 6 barg, max_T 150 C for K2750 and derived for streams 2750.001.6/*at N3*/model stream VENT_INT/FEED). No pump, exchanger area, tank residence, relief or utility supply ratings appear on this column-only P&ID extract; none added. Later alternatives must not raise these 6 barg/150 C ratings per instruction.